A 2D fixed-wing aircraft carries a "2D IMU" that measures body-frame angular velocity ω and body-frame acceleration $\mathbf{a} = (a_x, a_y)^T$. Your task is to propagate the aircraft's state (orientation, velocity, position) using the $\mathrm{SE}_2(2)$ group structure—a 2D analog of the 3D IMU navigation problem from the $\mathrm{SE}_2(3)$ lecture.